Adaptive Early Exit for Efficient Reasoning: Reproducing and Extending DEER

CMPE 258 Project Check-In & Initial Report Draft (Group 10)

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***Abstract*—Reasoning language models often generate extensively long chains of thought even after they have accumulated sufficient information to answer a question correctly. While recent approaches like Dynamic Early Exit in Reasoning Models (DEER) introduce a training-free mechanism to stop reasoning when confidence exceeds a fixed threshold, there is room for optimization. This project investigates a trajectory-aware adaptive stopping policy to improve the accuracy-computation tradeoff. Instead of relying solely on a fixed confidence threshold, our method trains a lightweight neural stopping controller using signals such as current confidence, confidence change across checkpoints, reasoning length, and checkpoint position. We will evaluate this adaptive approach against the standard DEER baseline and full Chain-of-Thought (CoT) decoding on the GSM8K and MATH-500 datasets. We anticipate that adaptive stopping will maintain accuracy while significantly reducing the number of reasoning tokens required across questions of varying difficulty.**

***Index Terms*—Chain-of-Thought, Early Exit, Large Language Models, Reasoning Efficiency, Dynamic Stopping.**

I. Introduction

The computational cost of reasoning in Large Language Models (LLMs) has increased dramatically with the adoption of long Chain-of-Thought (CoT) prompting and test-time compute scaling [2]. Reasoning models frequently overgenerate verification tokens even when a definitive answer has already been derived. The recently proposed Dynamic Early Exit in Reasoning Models (DEER) framework [1] addresses this inefficiency by periodically generating intermediate trial answers at reasoning checkpoints and halting generation once the model's confidence crosses a predefined static threshold.

In this work, we first reproduce the DEER methodology on a manageable open-source reasoning model and then extend it. We hypothesize that a single fixed confidence threshold is suboptimal across varied reasoning trajectories and problem difficulties. By developing a trajectory-aware adaptive stopping policy, we aim to dynamically assess intermediate answer reliability and achieve a superior accuracy-versus-compute tradeoff without sacrificing final answer accuracy.

II. Finalized Project Specifications

Following proposal feedback, the finalized specifications and technical pipeline for our system are structured into five core components:

A. System Pipeline and Experimental Steps

- **Baseline Reproduction:** We will set up an open-source reasoning model (e.g., from the `DeepSeek-R1-Distill` family [2]) and reproduce both the Full Chain-of-Thought (Full CoT) and standard DEER [1] baselines using the official implementation.
- **Fixed-Threshold Experiments:** We will evaluate standard DEER across a sweep of fixed confidence thresholds (e.g., $\tau \in \{0.60, 0.70, 0.80, 0.90, 0.95\}$) to establish baseline accuracy-computation curves on our target benchmarks.
- **Trajectory Feature Extraction:** At each candidate reasoning checkpoint k (exit opportunity), we will extract a trajectory feature vector $\mathbf{x}_k$ capturing intermediate trial-answer confidence c_k, confidence change across checkpoints ($\Delta c_k = c_k - c_{k-1}$), cumulative reasoning token length L_k, checkpoint position k, and trial-answer correctness.
- **Adaptive Early-Exit Controller:** Using the logged trajectory features, we will train a lightweight neural stopping controller (Multi-Layer Perceptron binary classifier) to dynamically predict whether an intermediate trial answer is sufficiently reliable to terminate reasoning.
- **Performance Evaluation:** We will compare Full CoT, fixed-threshold DEER, and Adaptive DEER under identical model and decoding settings. Key evaluation metrics include answer accuracy, average reasoning tokens, token-reduction ratio, total generated compute tokens, and inference latency.

B. Team Roles and Responsibilities

- **Jaswanthi Mandalapu (Baseline Reproduction & Model Inference):** Set up the reasoning model and

datasets, reproduce Full CoT and standard DEER baselines, and execute the fixed-threshold experiments.

- **Sai Chetan Muppalla (Adaptive Early-Exit Method):** Develop the trajectory-aware adaptive stopping approach, extract intermediate confidence and reasoning features, and implement/train the lightweight neural stopping controller.
- **Vishnu Sree Peram (Evaluation & Experimental Analysis):** Develop the evaluation pipeline, measure accuracy, reasoning-token usage, compute-token usage, and inference latency, and generate the accuracy-versus-compute tradeoff visualizations.

III. Data Sources and Exploration

A. Benchmark Datasets

The project relies on two primary, publicly available benchmarks containing grade-school and competition-level mathematical reasoning problems (summarized in Table I):

1) **GSM8K [3]:** A dataset of 8,792 high-quality, linguistically diverse grade-school math word problems (7,473 training and 1,319 test problems) requiring between 2 and 8 sequential steps to solve.
2) **MATH-500:** A challenging 500-problem subset of the competition-level Hendrycks MATH benchmark spanning seven mathematical subjects and five difficulty tiers (Levels 1–5).

TABLE I
SUMMARY OF DATASETS USED FOR TRAINING AND EVALUATION

Dataset	Split	Size	Difficulty & Format
GSM8K [3]	Train / Test	7,473 / 1,319	Grade-School (#### <num>)
MATH-500	Test	500	Competition Lvl 1–5 (\boxed{})

B. Current Exploration Status

We have located and initialized the Hugging Face data loaders for both GSM8K (`openai/gsm8k`) and MATH-500 (`HuggingFaceH4/MATH-500`). Our initial data exploration has focused on:

- **Prompt and Trace Formatting:** Inspecting problem lengths, multi-step solution chains, and delimiter tokens used to trigger intermediate trial-answer prompts in DEER.
- **Answer Parsing & Normalization:** Building regular-expression extraction utilities to parse final numerical answers following the #### delimiter in GSM8K and symbolic LaTeX expressions inside \boxed{...} in MATH-500, which is essential for labeling ground-truth correctness at intermediate exit checkpoints.

IV. Baseline Methodology: Dynamic Early Exit in Reasoning Models (Added as part of CheckIn-2) — Added by Jaswanthi Mandalapu

In this section, we consider the Dynamic Early Exit in Reasoning Models (DEER) framework [1], which aims to reduce the computational cost associated with long Chain-of-Thought (CoT) reasoning. We first describe the conventional reasoning process and subsequently introduce the intermediate trial-answer mechanism and the confidence-based stopping criterion employed in DEER.

A. Problem Formulation

Consider an autoregressive reasoning model f_θ, parameterized by θ, and an input problem x. Under conventional CoT decoding, the model generates a sequence of reasoning tokens

$$R = (r_1, r_2, \ldots, r_T), \tag{1}$$

followed by a final answer $\hat{y}$, where T denotes the length of the reasoning trajectory. The generation process continues until the model produces a termination token or reaches a prescribed generation limit.

Although additional reasoning steps may improve the reliability of the final answer, they also increase the computational cost of inference. Moreover, the number of reasoning tokens required to solve a problem depends on its difficulty and the reasoning trajectory followed by the model. Consequently, allowing the model to generate its complete reasoning trajectory may result in unnecessary computation, particularly when a reliable answer can be obtained at an earlier stage.

The objective of early-exit reasoning is to identify an intermediate point along the reasoning trajectory at which the model can terminate generation while preserving the correctness of its final answer.

B. Intermediate Trial-Answer Generation

DEER introduces intermediate exit opportunities at selected positions along the reasoning trajectory. Let

$$0 < t_1 < t_2 < \cdots < t_K \tag{2}$$

denote the sequence of reasoning checkpoints, where K is the number of candidate exit opportunities.

At checkpoint k, the model has generated a partial reasoning trajectory

$$R_k = (r_1, r_2, \ldots, r_{t_k}). \tag{3}$$

Conditioned on the input problem x and the partial reasoning trajectory R_k, the model is prompted to generate an intermediate trial answer, denoted by $\hat{y}_k$. We represent this operation as

$$\hat{y}_k = f_\theta(x, R_k). \tag{4}$$

The intermediate trial answer serves as an estimate of the final answer that the model could produce using the reasoning information available at checkpoint k.

The central idea is that a reasoning model may obtain sufficient information to solve a problem before completing its entire reasoning trajectory. By examining intermediate trial answers, DEER attempts to identify such instances and terminate the generation process without requiring additional reasoning tokens.

C. Confidence-Based Stopping Rule

We next describe the confidence-based stopping rule employed in DEER. Let c_k denote the confidence score associated with the trial answer $\hat{y}_k$ at checkpoint k. The confidence score is obtained using the confidence estimation procedure associated with the underlying reasoning model.

Given a prescribed confidence threshold τ, the stopping decision at checkpoint k is defined as

$$d_k = \begin{cases} 1, & c_k \geq \tau, \\ 0, & c_k < \tau. \end{cases} \tag{5}$$

Here, $d_k = 1$ indicates that the reasoning process is terminated and the corresponding trial answer is returned. Otherwise, the model continues generating reasoning tokens until the next checkpoint.

The stopping checkpoint is therefore given by

$$k^* = \inf \{k \in \{1, \ldots, K\} : c_k \geq \tau\}, \tag{6}$$

provided that the set is nonempty. If the stopping condition is not satisfied at any checkpoint, the model continues until its normal termination condition.

The choice of τ determines the tradeoff between computational efficiency and answer reliability. A smaller threshold permits earlier termination but may increase the probability of accepting an incorrect intermediate answer. Conversely, a larger threshold imposes a stricter confidence requirement and may result in longer reasoning trajectories.

D. Limitations of Fixed-Threshold Stopping

A key characteristic of the standard DEER framework is that the confidence threshold remains fixed across different input problems and reasoning checkpoints.

While this approach provides a simple stopping mechanism, it does not explicitly account for variations in problem difficulty or the evolution of confidence along a reasoning trajectory.

In particular, two reasoning trajectories may attain similar confidence scores at a given checkpoint, despite exhibiting substantially different confidence trends. For example, the confidence associated with one trajectory may increase consistently across successive checkpoints, whereas another trajectory may exhibit large fluctuations.

A stopping rule based exclusively on the current confidence score does not distinguish between these cases.

Furthermore, the reliability of an intermediate answer may depend on additional factors, including the number of reasoning tokens generated, the relative position of the checkpoint, and the variation in confidence between successive checkpoints.

These observations suggest that incorporating trajectory-dependent information into the stopping decision may improve the accuracy-computation tradeoff.

Motivated by this limitation, our project investigates an adaptive stopping mechanism in which the exit decision depends not only on the current confidence score but also on features extracted from the preceding reasoning trajectory.

E. Performance Metrics

We conclude this section by introducing the performance metrics used to evaluate the baseline DEER framework.

Consider a collection of N reasoning problems. Let A_i denote the indicator that the final answer produced for problem i is correct. The average answer accuracy is defined as

$$\mathrm{Acc} = \frac{1}{N} \sum_{i=1}^{N} A_i. \tag{7}$$

Let L_i^{CoT} and L_i^{DEER} denote the number of reasoning tokens generated for problem i under Full CoT decoding and DEER, respectively.

The corresponding average reasoning lengths are

$$\overline{L}_{\mathrm{CoT}} = \frac{1}{N} \sum_{i=1}^{N} L_i^{\mathrm{CoT}}, \tag{8}$$

and

$$\overline{L}_{\mathrm{DEER}} = \frac{1}{N} \sum_{i=1}^{N} L_i^{\mathrm{DEER}}. \tag{9}$$

We define the relative reduction in reasoning tokens as

$$\eta = 1 - \frac{\overline{L}_{\mathrm{DEER}}}{\overline{L}_{\mathrm{CoT}}}. \tag{10}$$

A larger value of η indicates a greater reduction in reasoning-token generation relative to Full CoT decoding.

However, the intermediate trial-answer generation and confidence estimation procedures introduce additional computational overhead. Therefore, reasoning-token reduction alone may not accurately characterize the overall inference efficiency.

To account for this overhead, we additionally consider the total number of generated tokens and the end-to-end inference latency.

Together, these metrics allow us to examine the tradeoff between answer accuracy and computational cost, and provide a common basis for comparing Full CoT decoding, standard DEER, and the proposed adaptive stopping framework.

V. Evaluation Methodology and Results

In this section, we present the experimental methodology and preliminary results obtained from reproducing the baseline DEER framework. We first compare conventional Full Chain-of-Thought (CoT) decoding with confidence-based early-exit reasoning on a subset of the GSM8K benchmark. We subsequently examine the accuracy-computation tradeoff, identify limitations of the current experimental setup, and outline the planned extensions.

A. Jaswanthi Mandalapu: Baseline Reproduction and Evaluation

1) Experimental Methodology: We consider the problem of reducing inference-time computation in reasoning language models while preserving the correctness of the generated answers. To this end, we conducted an initial evaluation of

conventional Full CoT decoding and the DEER framework [1] on the GSM8K benchmark [3]. The preliminary evaluation was performed on a subset of 20 mathematical reasoning problems. For each problem, we obtained the final answer and the corresponding token usage under both inference strategies. Under Full CoT decoding, the model was allowed to generate its reasoning trajectory until normal completion. In contrast, DEER introduced intermediate exit opportunities during reasoning and employed a confidence-based stopping criterion to determine whether additional reasoning was necessary. We evaluated the two inference strategies using the following metrics:

1) *Answer Accuracy:* The fraction of problems for which the generated final answer matches the ground-truth answer.
2) *Average Token Usage:* The average number of generated tokens across the evaluated problems.
3) *Question-wise Token Usage:* The number of generated tokens for each problem under the two inference strategies.
4) *Relative Token Reduction:* The percentage reduction in average generated tokens obtained using DEER relative to Full CoT.

The purpose of this preliminary experiment is to establish a baseline for subsequent evaluations involving larger datasets and adaptive stopping mechanisms.

2) Accuracy Comparison: We first examine the answer accuracy obtained under Full CoT and DEER. The results are summarized in Table II.

TABLE II
PRELIMINARY RESULTS ON 20 GSM8K PROBLEMS

Metric	Full CoT	DEER
Accuracy (%)	65.0	70.0
Correct Answers	13	14
Average Tokens	634.5	600.5

We observe that Full CoT achieves an accuracy of 65%, corresponding to 13 correctly solved problems, whereas DEER achieves an accuracy of 70%, corresponding to 14 correctly solved problems. Thus, DEER achieves a five-percentage-point improvement in accuracy on the evaluated subset. This observation suggests that early termination does not necessarily degrade answer correctness. In certain instances, additional reasoning may alter an otherwise correct intermediate answer. However, since the evaluation consists of only 20 problems, the observed difference corresponds to one additional correct answer. Consequently, the present results should be interpreted as preliminary rather than as evidence of a systematic accuracy improvement.

3) Average Token Usage: We next compare the average number of generated tokens under the two inference strategies. As shown in Fig. 2, Full CoT generates an average of 634.5 tokens per problem, whereas DEER generates 600.5 tokens. The relative reduction in average token usage is given by

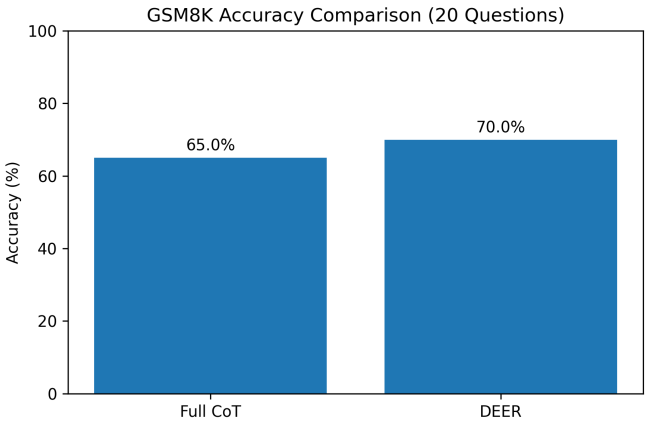


Fig. 1. Answer accuracy of Full CoT and DEER on 20 GSM8K problems.

$$\eta = \left(1 - \frac{600.5}{634.5}\right) \times 100\% \approx 5.36\%. \tag{11}$$

Thus, DEER achieves an average token reduction of approximately 5.36% on the evaluated subset. Although the reduction is modest, it indicates that intermediate stopping decisions can reduce token generation without an observed loss in aggregate answer accuracy.

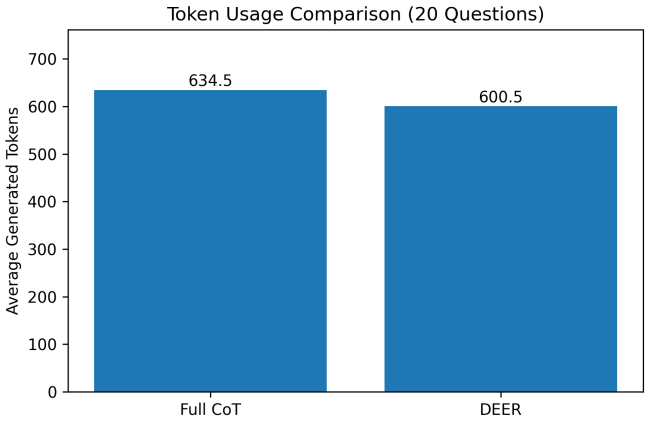


Fig. 2. Average generated tokens under Full CoT and DEER on 20 GSM8K problems.

4) Question-wise Token Usage: To further examine the behavior of DEER, we compare the number of generated tokens for individual problems. Fig. 3 presents the question-wise token usage under Full CoT and DEER, with problems ordered according to their Full CoT token counts. We observe substantial variation in the token savings across problems. In particular, DEER reduces token usage significantly for certain long reasoning trajectories, whereas the savings are comparatively small for several shorter trajectories. Moreover, DEER occasionally generates more tokens than Full CoT. This behavior indicates that intermediate trial-answer generation and the stopping mechanism may introduce additional computational

overhead. These observations suggest that the effectiveness of early-exit reasoning depends on the characteristics of the individual reasoning trajectory.

5) Analysis of Long Reasoning Trajectories: We further investigate the behavior of DEER on three representative problems with relatively long Full CoT reasoning trajectories. Fig. 4 compares the token usage for questions Q8, Q19, and Q13. For Q19, DEER reduces the token count from approximately 2040 to 720, corresponding to a reduction of approximately 65%. For Q8, the reduction is comparatively smaller, with the token count decreasing from approximately 2040 to 1750. Similarly, for Q13, the token count decreases from approximately 1210 to 790. These results indicate that the computational savings obtained through early termination are not uniform across problems. In particular, the substantial reduction observed for Q19 suggests that the model may generate a reliable intermediate answer considerably earlier than the completion of its full reasoning trajectory. However, a long reasoning trajectory does not necessarily imply that an early-exit mechanism will achieve a large reduction in token usage. The observed variation motivates a more detailed investigation of the relationship between intermediate answer confidence, reasoning length, and stopping decisions.

6) Discussion of Preliminary Findings: The preliminary evaluation provides three observations. First, DEER achieves comparable, and in this experiment slightly higher, answer accuracy than Full CoT while reducing the average number of generated tokens. Second, the computational savings vary considerably across individual problems. In particular, some long reasoning trajectories exhibit substantial token reductions, whereas others provide only limited savings. Third, DEER does not consistently reduce token usage for every problem. In some instances, the additional computation associated with intermediate trial answers may outweigh the savings obtained through early termination. These observations highlight the importance of evaluating early-exit reasoning at the level of individual trajectories rather than relying exclusively on aggregate accuracy and token-reduction metrics.

7) Limitations of the Current Experimental Setup: The present evaluation is preliminary and has several limitations. First, the experiments were conducted on only 20 GSM8K problems. This sample size is insufficient to establish statistically reliable conclusions regarding the accuracy-computation tradeoff. Second, the current evaluation is restricted to GSM8K and does not include more challenging mathematical reasoning benchmarks, such as MATH-500. Third, the aggregate token statistics do not distinguish between tokens generated during the main reasoning trajectory and those generated during intermediate trial-answer evaluation. Consequently, the present results do not fully characterize the computational overhead introduced by DEER. Fourth, the current plots do not provide a detailed analysis of the intermediate confidence scores or the checkpoints at which reasoning terminates. Finally, the results correspond to an initial baseline evaluation. A systematic comparison across multiple confidence thresholds and decoding configurations remains necessary.

8) Planned Extensions: Motivated by the preliminary observations, we plan to extend the evaluation along the following directions.

1) *Larger-Scale Evaluation:* Extend the experiments to the complete GSM8K test set and subsequently evaluate the baseline on MATH-500.
2) *Confidence Threshold Analysis:* Evaluate DEER across multiple fixed confidence thresholds to characterize the tradeoff between answer accuracy and token usage.
3) *Checkpoint-Level Analysis:* Record intermediate trial answers, confidence scores, checkpoint positions, and stopping decisions to understand how early-exit behavior varies across reasoning trajectories.
4) *Computational Overhead Analysis:* Separately measure reasoning tokens, trial-answer tokens, total generated tokens, and inference latency.
5) *Adaptive DEER Comparison:* Compare the fixed-threshold baseline with the proposed trajectory-aware stopping mechanism to determine whether adaptive stopping can provide a more favorable accuracy-computation tradeoff.

The objective of these extensions is to obtain a more comprehensive characterization of early-exit reasoning and establish a reliable experimental baseline for evaluating the proposed adaptive stopping framework.

VI. Project Management and Workspace

We have established our collaborative workspace to manage source code, dataset scripts, experiment logs, and report documentation:

- **GitHub Repository:** Initialized for version control of baseline reproduction and controller scripts: https://github.com/PeramVishnuSree/Adaptive-Early-Exit-for-Efficient-Reasoning-Reproducing-and-Extending-DEER
- **Google Drive Folder:** Initialized for storing model checkpoint logs, trajectory datasets, and report drafts: https://drive.google.com/drive/folders/1F4p6y-iFyPqzfq8yA-skQIf8niTviEIs?usp=drive_link
- **Instructional Team Access:** We have invited the instructor and TA to our project workspace using the required email addresses.
- **Computing Resources:** We plan to utilize the SJSU College of Engineering HPC cluster alongside cloud GPU environments for model inference and trajectory collection.

VII. Tentative Outline for Final Report

To align with IEEE conference formatting requirements, our final report will be organized into the following sections:

- **Abstract:** Summary of the problem, proposed trajectory-aware adaptive stopping policy, and key empirical findings.
- **Section I — Introduction:** Motivation, test-time compute overthinking in reasoning LLMs, limitations of static early-exit thresholds, and project objectives.

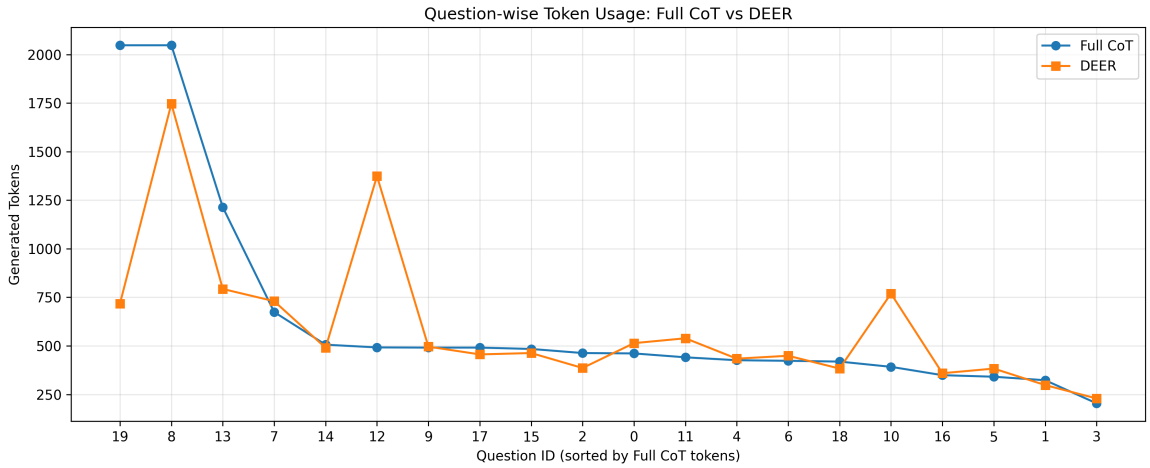


Fig. 3. Question-wise generated token comparison between Full CoT and DEER. Questions are ordered according to their Full CoT token usage.

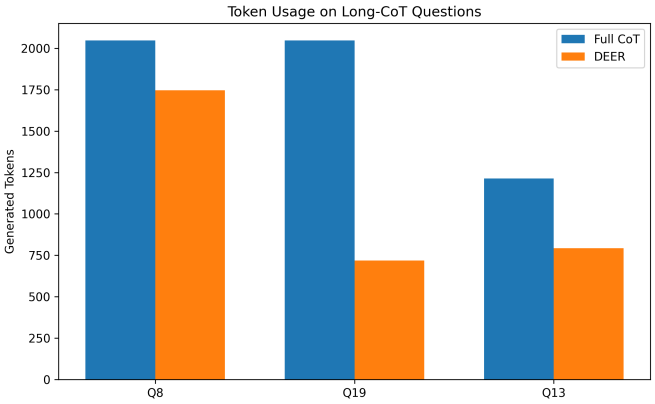


Fig. 4. Token usage comparison for three representative long-CoT problems.

- **Section II — Related Work:** Literature review of reasoning LLMs [2], Chain-of-Thought prompting, and dynamic early-exit mechanisms [1].
- **Section III — Datasets and Preprocessing:** Detailed breakdown of GSM8K [3] and MATH-500 splits, answer extraction, and trajectory dataset construction.
- **Section IV — Methodology:**
  - *A. Baseline Formulation:* Reproducing Full CoT and DEER with static confidence thresholds.
  - *B. Trajectory Feature Engineering:* Extraction of confidence level, confidence delta, token length, and checkpoint position.
  - *C. Proposed Adaptive Controller:* Architecture and training objective of the lightweight neural stopping policy.
- **Section V — Experimental Setup:** Open-source model specifications, decoding hyperparameters, hardware setup, and evaluation metrics.
- **Section VI — Results and Discussion:**
  - *A. Accuracy vs. Reasoning-Token Tradeoff Curves.*
  - *B. Fixed-Threshold vs. Adaptive DEER Across Difficulty Levels.*
  - *C. Feature Ablation, Compute Overhead, and Error Analysis.*
- **Section VII — Conclusion and Future Work:** Summary of contributions, limitations, and potential future extensions.

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